Technical Document #958: Cloud Computing - Comprehensive Overview

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Infrastructure as Code (IaC) manages infrastructure using machine-readable configuration files. Terraform and CloudFormation are popular IaC tools.

Microservices architecture structures applications as a collection of loosely coupled services. Each service runs in its own process and communicates via APIs.

Multi-cloud strategies use services from multiple cloud providers. They reduce vendor lock-in and optimize for cost and performance.

Kubernetes orchestrates containerized applications across clusters of machines. It handles deployment, scaling, and management of containerized workloads.

Content delivery networks (CDNs) distribute content globally to reduce latency. CloudFront and Cloudflare are popular CDN providers.

The rapid advancement of technology has transformed how organizations approach problem-solving and innovation. Companies must adapt to stay competitive in an ever-evolving landscape.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Key Takeaways:

- Platform as a Service (PaaS) provides a platform for developing and deploying applications.
- Containerization packages applications with their dependencies.
- Infrastructure as Code (IaC) manages infrastructure using machine-readable configuration files.